Coding / Historical Gpt Gemini Comparison

PDF edition of preserved repository document | docs/coding/historical_gpt_gemini_comparison.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Historical HumanEval references

These are historical report values, not newly rerun API models. Our scores use the same benchmark task family but a different instruction prompt and CPython3.14.7 WASI runtime. Neither sampling nor contamination control is matched sufficiently to establish general superiority.

Historical model	Reported HumanEval score	Reported prompting	Primary source
GPT-3.5	48.1%	0-shot	[GPT-4 technical report, Table2](https://cdn.openai.com/papers/gpt-4.pdf)
GPT-4	67.0%	0-shot	[GPT-4 technical report, Table2](https://cdn.openai.com/papers/gpt-4.pdf)
Gemini1.0 Pro	67.7%	0-shot, instruction-tuned	[Gemini report v1, Table2 p7](https://arxiv.org/pdf/2312.11805v1)
Gemini1.0 Ultra	74.4%	0-shot, instruction-tuned	[Gemini report v1, Table2 p7](https://arxiv.org/pdf/2312.11805v1)

The GPT report was initially published March2023; exact API checkpoint identifiers and complete HumanEval decoding/sample settings are not specified in that table. The Gemini report v1 is December2023, with table API comparisons collected November2023. Its instruction-tuned flag does not supply a complete replicable prompt/decoding protocol. No temperature, number of samples, parameter count or hidden model revision is inferred from omitted information. Values are treated as published functional-correctness percentages; our measured statistic is one-completion pass@1, and raw numerical differences are descriptive only.

The pinned Gemini v1 PDF SHA-256 is f3c5b175d26159b92891b32380515ec1036f4ffe2862e19d7277fd165e0d9663. A current hosted report has later revisions, so v1 is used for historical attribution. The official HumanEval repository (https://github.com/openai/human-eval) supplies the164 tasks. Pretraining contamination of our open models is unknown. Local split disjointness cannot resolve this limitation.

The earlier0.5B open-model run remains94/164,57.317073% under its recorded local greedy512-token protocol. The final measured1.5B run is reported separately in the machine scorecard after completion. Claims such as “beats GPT-4,” “contamination-free” or “general coding perfection” are not supported by this comparison.